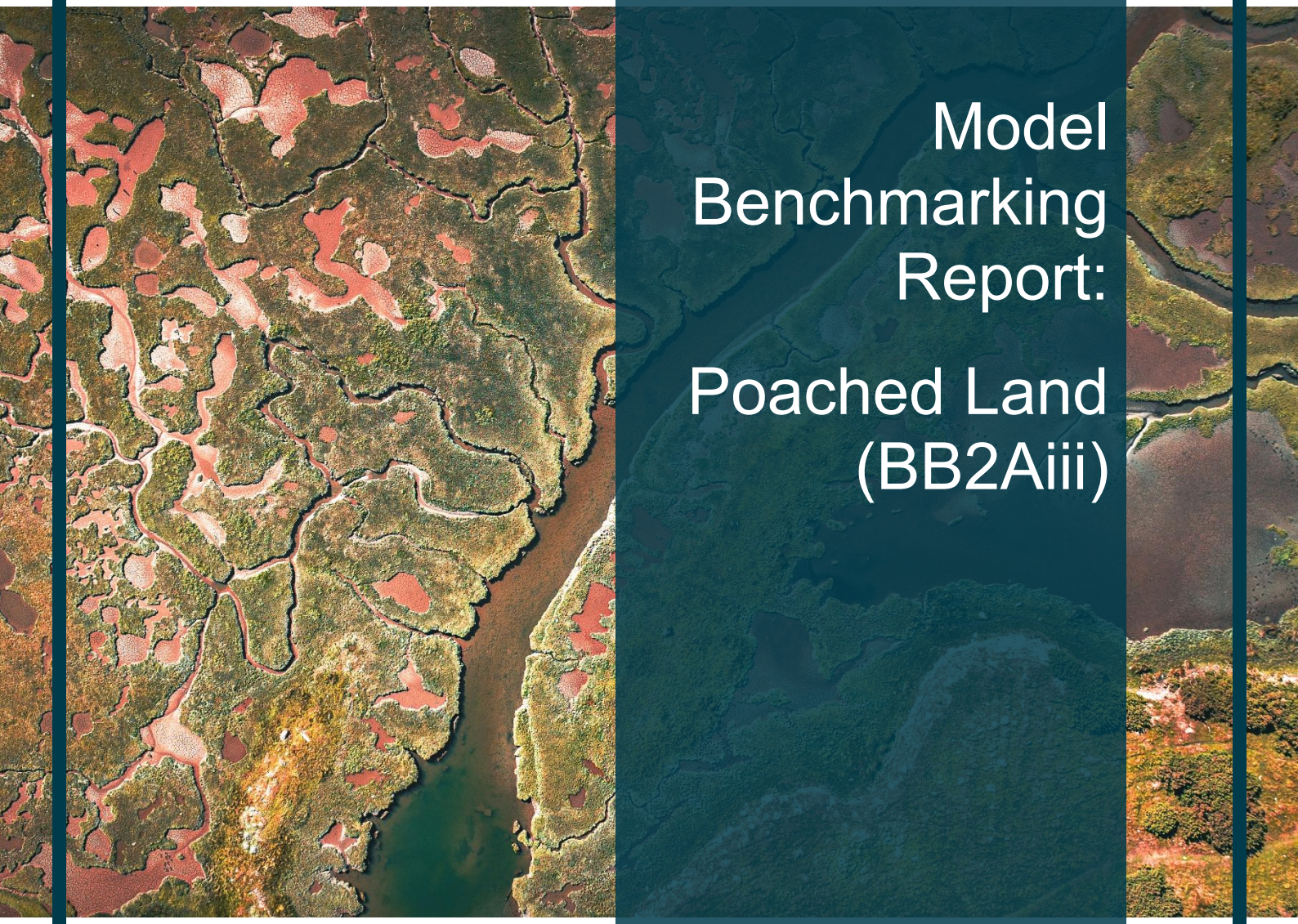
River Deep Mountain AI



Model Benchmarking Report:

Poached Land (BB2Aiii)

ADAS

February 2026

Model Benchmarking Report - Poached Land (BB2Aiii)

FINAL REPORT

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Glossary

ALERT	Agricultural Land Environmental Risk and Opportunity Tool
BSI	Bare Soil index
GEE	Google Earth Engine
MAGIC	Multi-Agency Geographic Information for the Countryside River
NDVI	Normalized Difference Vegetation Index
RDMAI	River Deep Mountain AI
S2DR3	Sentinel 2 Deep Resolution 3.0
WFD	Water Framework Directive

Summary

Table 1.1 Benchmarking summary

Model Outputs	Manual review of data / catchment walkover	RDMAI Model
Model Approach	Manual review of available datasets and in-person catchment walkover to identify features of interest.	Satellite imagery of 1m resolution is uploaded into Google Earth Engine (GEE), along with mask data derived from OSM. The model is then run in GEE, identifying poached land based on spectral data and derived vegetation indices.
Applications	Identify areas of high risk for agricultural pollution and inform where placement of mitigation measures would be beneficial.	Identify areas of high risk for agricultural pollution to help target catchment walkovers and inform where placement of mitigation measures would be beneficial.
Geographic Constraints	Theoretically none, although land access is an issue. Not all potential areas of interest are visible from roads, footpaths and other publicly accessible spaces.	The model can be applied anywhere Sentinel-2 imagery is available, but is currently tested only in the UK. Accuracy depends on training data, lighting conditions, and regional differences; so new regions require new training polygons.
Temporal Resolution	Catchment walkovers can be undertaken as frequently as desired (given practical and cost considerations). A manual search for poached land undertaken solely using imagery data freely available on the internet would be constrained by the dates and frequency of such imagery.	The model is event- or date-specific, running on a single Sentinel-2 acquisition. The temporal resolution depends on available Sentinel-2 imagery with sufficiently low cloud cover for the area of interest.

Data Inputs Required	A manual search for poached land can be undertaken using imagery data freely available on the internet such as that included in Google Maps.	The model requires pre-processed S2DR3 imagery uploaded as assets in GEE, along with user-provided training polygons for landcover classes.
Model Performance	It is not possible to quantitively assess performance Performance is constrained by the ability to locate and map all poached areas, given accessibility and time constraints.	It is not possible to quantitively assess model performance. However, the model appears to over predict poaching, and misses significant areas where poaching is visible, particularly in the Northumbrian catchment.
Cost & Time	A thorough review of data for a small (~ 10 km^2) catchment would take a few hours, a fairly rapid catchment walkover of a similar area would take around a day. Only significant costs are staff time to undertake the work.	Preparing the input data takes roughly an hour. After that, running the JavaScript model in GEE takes approximately 15 minutes per catchment. The S2DR3 imagery requires a licence for commercial use.
Strengths	A walkover provides more detailed insight into the severity of any poached land, the connectivity of that land and the overall risk to water quality.	Simple to use once familiar with the model. Offers repeatability and consistency between users.
Limitations	Labour intensive. Results are subjective and depend upon the user and the time available for the task.	Requires training data, increasing upfront effort and costs when deploying in new areas or with new users. Availability of cloud-free imagery Accuracy of the S2DR3 downscaled imagery

1. Benchmarking Methodology

1.1 Model Selection

1.1.1 RDMAI Poached Land Detection Model

Poached land, caused by livestock and farm machinery, poses a greater risk of runoff and erosion. The Open Poached Land Detection Model identifies these areas of degraded soils, to aid the targeting of catchment walkovers and the placement of agricultural pollution control measures. The model uses a hybrid approach that integrates AI-upscaled satellite imagery, machine learning algorithms, remote sensing indices, and GIS-based spatial analysis. Sentinel-2 imagery, at 10- m resolution is downscaled to 1-m using a Sentinel 2 Deep Resolution 3.0 (S2DR3) tool. A random forest classifier is then applied to classify land cover and detect poached areas based on spectral and spatial patterns. Remote sensing indices, including NDVI and BSI, are used to extract key surface characteristics related to vegetation health and soil exposure. GIS processing is then used to remove isolated pixels and large field-scale areas.

1.1.2 Existing Industry Model(s)/Tool(s)

There is no existing model or dataset that shows the locations of poached land. Benchmarking has instead been against a catchment walkover, which would be used to identify a range of potential pollutant issues arising from agriculture, including those associated with poaching from livestock and farm machinery use.

1.2 Catchment Selection

Catchment walkovers were undertaken in two catchments (Figure 1-1 and Figure 1-2) in different parts of the country.

GB110060029250 (Fernhill Brook - headwaters to tidal limit) in the Carmarthen Bay and the Gower catchment. The area was known to have a number of dairy farms and therefore highly likely to have slurry tanks, bare ground and poached land. The catchment size (11 km^2) was appropriate for a walk over in a day.

GB103024077330 (Smallhope Burn from Source to Browney, Stocke Water Body) – a predominantly grassland catchment with cattle and sheep farms. The catchment has poor ecological status, in part due to phosphates and agriculture. The catchment size of 42 km^2 meant that only the lower half of the catchment was covered in detail by the walkover.

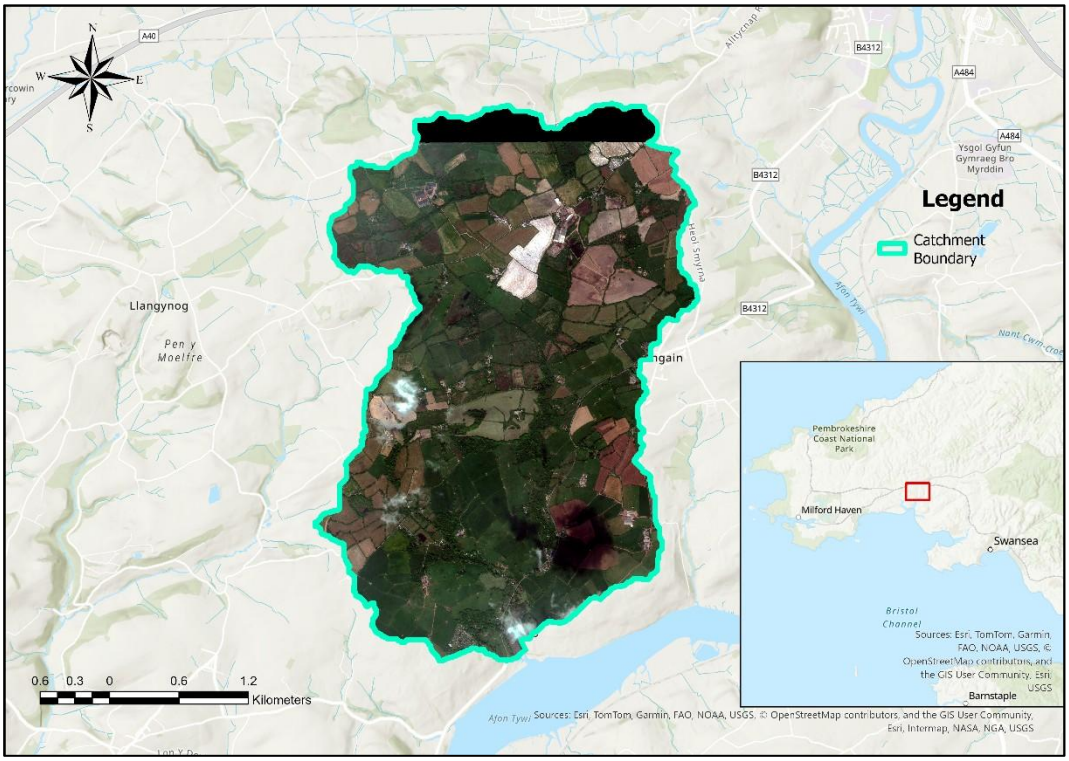

Figure 1-1 GB110060029250 catchment (Fernhill Brook - headwaters to tidal limit)

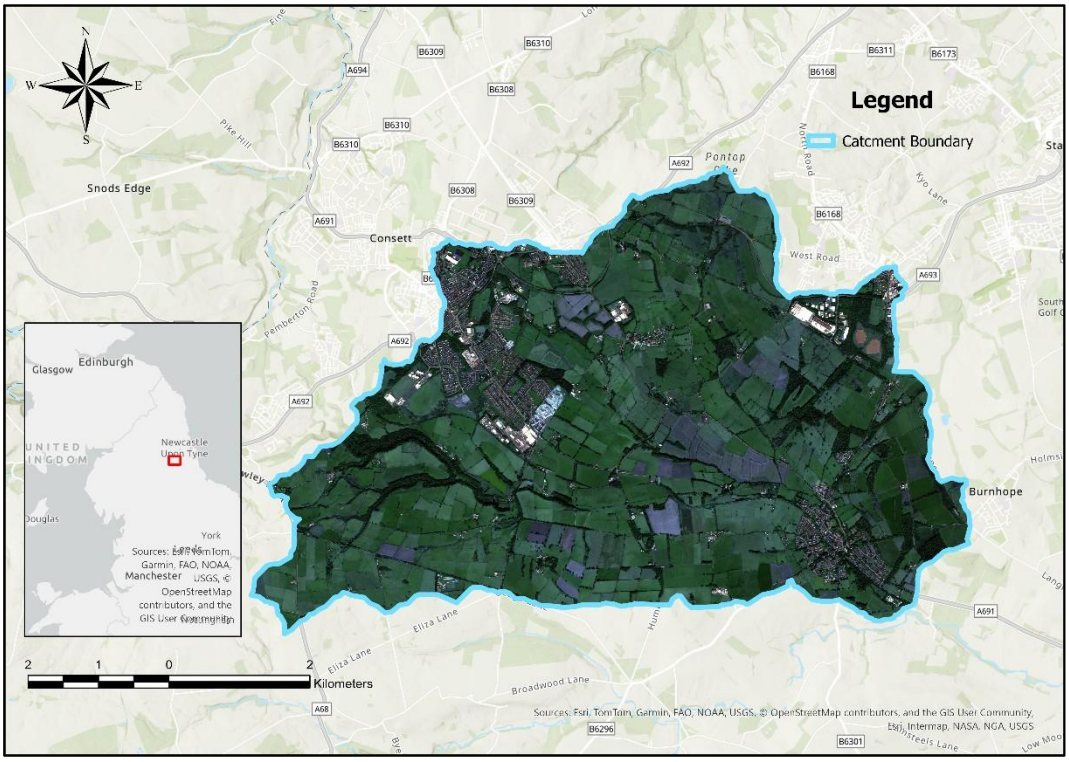

Figure 1-2 GB103024077330 (Smallhope Burn from Source to Browney, Stocke Water Body)

1.3 Benchmarking Methodology

The benchmarking compares the extent and location of poaching identified using the model with that identified from the walkovers of the catchments.

Prior to undertaking the actual walkover, the consultant carried out the series of checks that would normally be run through to identify potential locations of interest, high risk areas for diffuse pollution and any accessibility issues. The tools used for this include:

1. Agricultural Land Environmental Risk and Opportunity Tool (ALERT), which contains data layers such as land use, topography and satellite imagery, and was designed to help locate interventions to reduce pollution from agriculture

2. MAGIC in England or DataMapWales, which contain geographic information about the natural environment from across government, to understand the water quality status and pollution issues within the catchments

2. LandIS to identify soils, as pollutant issues can vary by soil type (e.g. lighter texture soils will have a higher nitrate leaching risk than heavier soils, heavier soils are more prone to waterlogging).

4. Google Earth imagery to inspect the catchment and identify areas that should be targeted for inspection (if safe and legal to do so). Date of most recent imagery noted and used, although historic imagery can also be utilised.

Reviewing these datasets for the catchments took several hours. The walkover was then undertaken, identifying the locations and features of interest, with access constrained to roads, footpaths and bridleways and thus only certain features being visible. The walkover was restricted in time to approximately one day, a reasonable time allowance for a first assessment with no land owner interactions. After the walkover was completed, the information gathered was documented and mapped.

1.4 Benchmarking Limitations

Due to access issues and weather (flooding) not all areas could be walked over or assessed. Areas of poaching are not always visible from public roads and footpaths.

The walkovers were undertaken in November and December 2025, whilst the Sentinel 2 imagery used for the RDMAI model was from September 2025.

Whilst the RDMAI model is repeatable and would produce consistent results, identification of features as part of a catchment walkover would vary depending upon the person undertaking the task, the time available and other factors. The assessment of model accuracy is thus subjective.

Given the format of the data produced and subjective nature of the walkover output, it is not possible to produce any quantitative assessment of model performance.

2. Benchmarking Results

2.1 Model 'Ease-of-Use'

Table 2.1 Comparison of model 'ease-of-use'

Model Outputs	Manual review of data / catchment walkover	RDMAI Model
Installation & Run Process	A catchment walkover can be undertaken by anyone with a reasonable understanding of agriculture and diffuse pollution. A more qualified individual, or one familiar with the catchment would speed up the process or provide more detailed insight. Land access is an issue. Not all potential areas of interest are visible from roads, footpaths and other publicly accessible spaces. Someone familiar with the catchment and local landowners may be able to arrange access to areas of private land. These walkovers were restricted to roads, footpaths and bridleways. The catchments visited in this project were approximately 11 km^2 and 40 km^2 in size, and it was not possible to cover the larger catchment in a day, even allowing for some preparatory work to identify areas of interest. The time taken also depends upon the level of detail being recorded. A risk assessment was undertaken prior to the walkover, this included lone working – which whilst avoidable would increase the staffing costs in order to be avoided.	While the core model runs in Google Earth Engine (GEE), several preparatory steps outside of GEE are also required. Specifically, the high-resolution S2DR3 10-band 1 m imagery must be generated using a Jupyter Notebook script. This script processes Sentinel-2 imagery and outputs the resulting files to Google Drive, from where they must be manually downloaded and then uploaded into GEE as assets. The workflow can be time-consuming, as it also requires selecting a suitable satellite acquisition date with low cloud coverage. Overall, this multi-step process demands familiarity with both Python/Jupyter environments and GEE, making it less straightforward than the other models. This model also requires users to provide training data. The training samples must be based on the S2DR3 10-band 1 m imagery selected, as regional differences, lighting conditions, and acquisition angles affect detection accuracy. Users must digitize features and assign a class number corresponding to landcover type: • 0 – Poached Land • 1 – Bare Cropland

		• 2 – Cropland • 3 – Trees • 4 – Waterbodies ADAS were provided with a training file so did not have to undertake this task. Finally, the poached-land-detection-model.js script must be updated with the correct GEE asset paths, acquisition dates, and the field name corresponding to landcover classes. Once all inputs are correctly configured, running the script in GEE is fast, typically taking less than two minutes to classify the full 1 m imagery.
Data inputs	A manual search for poached land can be undertaken using imagery data freely available on the internet such as that included in Google Maps.	The model requires pre-processed S2DR3 1 m imagery uploaded as assets in GEE, along with user-provided training polygons for landcover classes. The model handles most preprocessing tasks, including cloud filtering, built-up area masking, and compositing. Scripts automate processing, so minimal manual pre-processing is needed. Additional layers, such as roads or NDVI, are optional but recommended for post-classification refinement. The roads layer is derived from OpenStreetMap data and buffered by 3 m to act as a mask, helping to exclude road areas from poached land detection.
Licensing Requirements	None.	A Google Earth Engine (GEE) account is required to run the model. GEE operates on a “pay-as-you-go” system, where users are charged only for the computational credits used. For small study areas, this is very cost-effective. During testing, GEE provided an initial free credit allocation of $250. Running the model for the Welsh catchment used a

		negligible amount of credits, less than $0.01 per run. No additional licenses or paid software are required. The S2DR3 1m imagery requires a licence for commercial use, costing 2.4 euros per km^2 processed (with cheaper rates for much larger areas.

2.2 Model Outputs

Table 2.2 Comparison of model outputs

Model Outputs	Manual review of data / catchment walkover	RDMAI Model
Data Format	Outputs were as annotated maps and notes, subsequently converted to GIS shapefiles.	The model outputs classified landcover layers in vector format (shapefile), representing bare cropland polygons. Raster layers (e.g., BSI, NDVI) can also be exported optionally, but the main output is vector.
Temporal Resolution	Representative of the day of the walkover or the day any imagery used was produced. Whilst more walkovers could be repeated to observe change over time, there are financial and practical constraints given the time required.	The model is event- or date-specific, running on a single Sentinel-2 acquisition. The temporal resolution depends on available Sentinel-2 imagery with <20% cloud cover for the area of interest. It does not automatically produce multi-date time series.

Geographic Constraints	None. Although land access is an issue, and poached areas are not always visible from publicly available spaces.	The model can be applied anywhere Sentinel-2 imagery is available, but is currently tested only in the UK. Accuracy depends on training data, lighting conditions, and regional differences; so new regions require new training polygons. No inherent software restrictions exist beyond GEE access.

2.3 Time and Cost Comparison

A fairly rapid catchment walkover for a small (~10 km2) catchment can be undertaken in one day, plus some additional time for preparation (e.g. reviewing datasets) and reporting.

Preparing the input data to upload to GEE takes roughly an hour. After that, running the JavaScript model in GEE takes approximately 15 minutes per catchment. Once a user is familiar with the RDMAI model, it would offer significant time savings compared to manual identification. However, the model requires a moderate level of technical expertise, including Python environment setup, package version management, and basic image pre-processing. GEE is a paid for service, based on processing time, but the costs for running the benchmarking were found to be very small. The S2DR3 1m imagery requires a licence for commercial use.

Overall, the RDMAI model could reduce labour time spent on surveying large areas for poached, but still requires manual verification and interpretation to determine severity and the overall risk to water quality.

In reality however, the poached land identification model would not be used in isolation, and would form part of a toolset (alongside other models from RDMAI and the data listed in Section 1.3), to help information any catchment walkovers undertaken, rather than as an alternative to catchment walkovers.

2.4 Model Performance

For the Welsh catchment, the walkover identified a number of areas in the south of the catchment where poaching was observed on grassland fields (Figure 2-1). These areas did not always include features identified by the model. Towards the north of the catchment, some large fields were flagged as being poached because they had recently been reseeded following maize, and the wheelings and soil issues associated with maize growing could still be observed through the young grass – these features could be a potential pollutant source this winter, but much less so once the grass is more established. It is unlikely that such features would be identified by the RDMAI model (or a walkover undertaken at a different time) due to their transient nature and the accuracy of the crop mask used.

Figure 2-2 and Figure 2-3 show more detail for two areas identified by both the model and the walkover. The RDMAI model found the areas of poaching around the field entrances in the centre of the image in Figure 2-2, that were also picked up us part of the walkover. However, this image also shows that the model did not find the slightly smaller area of poaching to the top of the image, and also that it stated that large amounts of hedgerows in the top right were also poached, along with areas of healthy grassland in the bottom right. Figure 2-3 shows both approaches identified a large area of bare ground, although the RDMAI model did not include the central area, and extended the area downwards into what appears to be grassland.

Figure 2-4 shows the comparison between the RDMAI model and the walkover for the Northumbrian catchment. The model identified a lot of areas of land as poached, but these were frequently found to be areas within grassland fields or along hedgerows with no obvious signs

in the imagery of poaching. Figure 2-5 to Figure 2-8 show the satellite imagery for different areas where poaching was found as part of the walkover, but not by the RDMAI model, along with photographs of the areas taken during the walkover. Although the satellite imagery was from September and the walkover was undertaken in December, and so any poaching may not be as severe, the areas are still visible in the satellite imagery. Whilst the model does not identify these areas from the walkover as poaching, these four figures all show the model identifying areas of nearby healthy grassland as being poached (as was also found for the Welsh catchment). One possible reason for the omission in Figure 2-5 is that much of the field corner is shaded, although some of the poaching is still visible in the imagery.

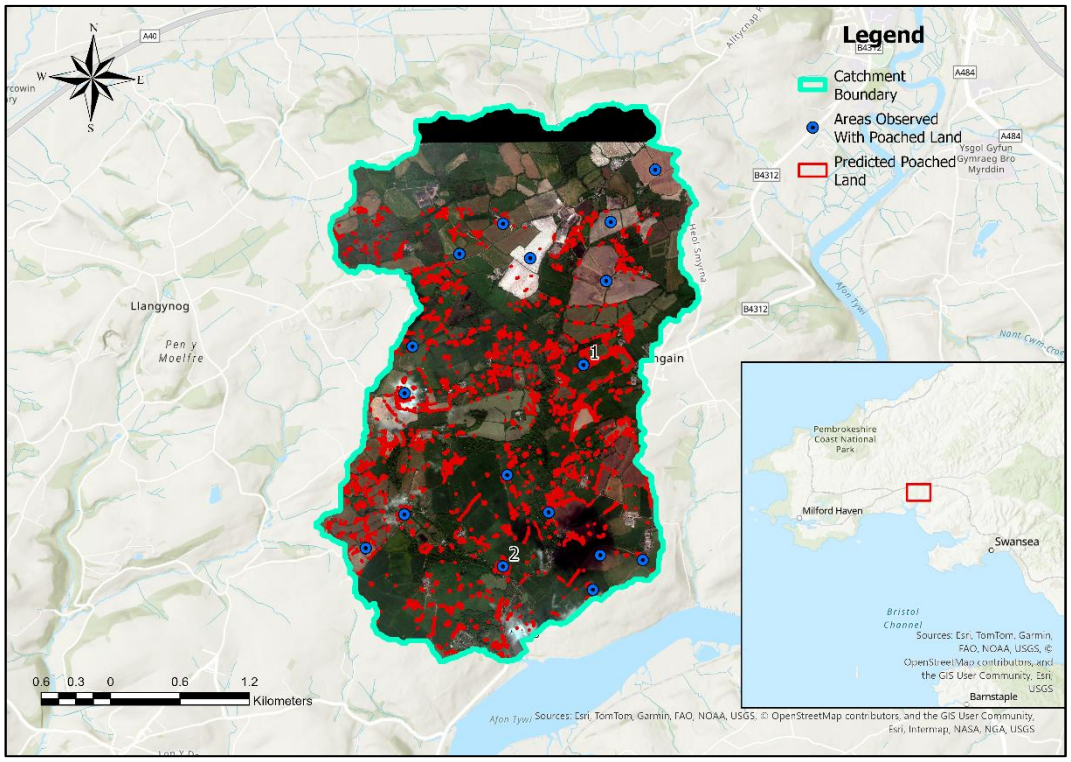

Figure 2-1 Comparison of poached land identified by the RDMAI model and the catchment walkover for the Welsh catchment (GB110060029250)

Figure 2-2 Poaching identified by the RDMAI model and the walkover at location 1 from Figure 2-1

Figure 2-3 Poaching identified by the RDMAI model and the walkover at location 2 from Figure 2-1

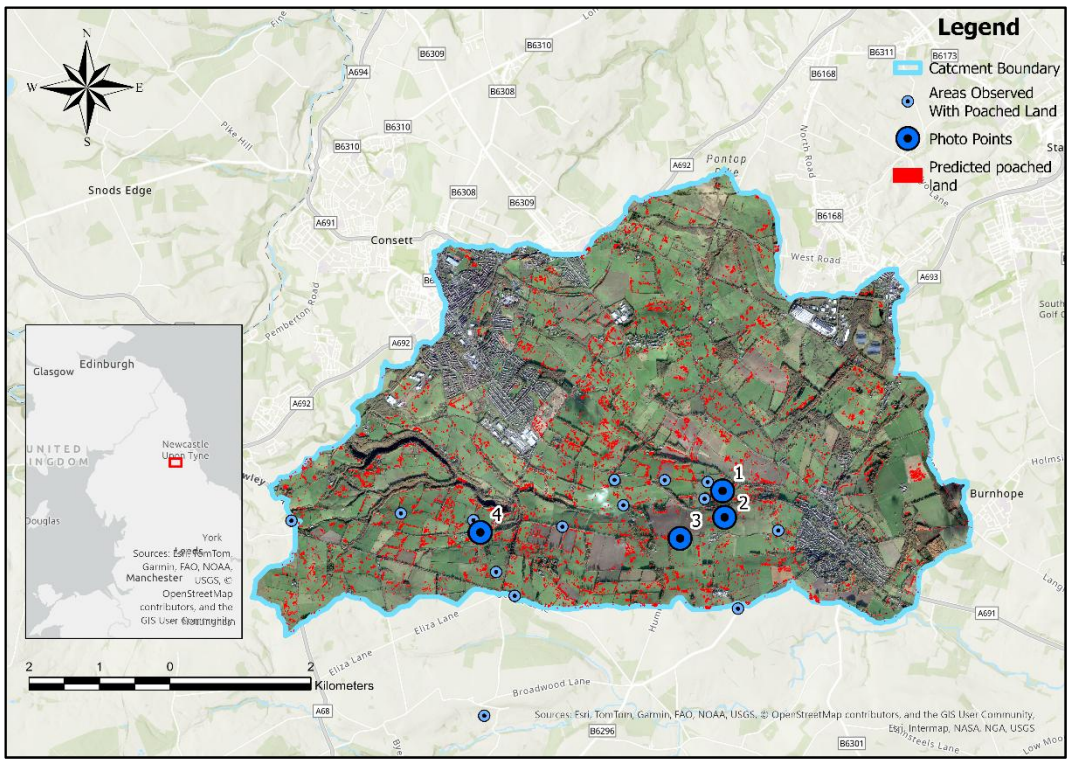

Figure 2-4 Comparison of poached land identified by the RDMAI model and the catchment walkover for the Northumbrian catchment (GB103024077330)

Figure 2-5 Photograph of poaching found by the catchment walkover for location 1, and a close-up of this area from satellite image used in the RDMAI model, also showing the poached areas identified by the model.

Figure 2-6 Photograph of poaching found by the catchment walkover for location 2, and a close-up of this area from satellite image used in the RDMAI model, also showing the poached areas identified by the model.

Figure 2-7 Photograph of poaching found by the catchment walkover for location 3, and a close-up of this area from satellite image used in the RDMAI model, also showing the poached areas identified by the model.

Figure 2-8 Photograph of poaching found by the catchment walkover for location 4, and a close-up of this area from satellite image used in the RDMAI model, also showing the poached areas identified by the model.

3. Benchmarking Conclusion

3.1 Summary

A summary table of key model characteristics can be found in Table 1.1, at the start of the report.

3.2 Conclusion

Whilst the model has potential, it is likely that further work is required to make it operationally useful. The model fails to detect areas of poaching and mis-classifies significant areas that are not poached. Even aside from the accuracy issues, given the wide extent and large number of areas identified as poaching in both catchments, the current iteration of the model would not enable a more-focused catchment walkover. As part of any future development, it would be worthwhile considering whether any prediction confidence or poaching severity metric could also be included in the model, as tweaking this could allow for improved performance or facilitate targeting for a catchment walkover.

Performance in the Northumbrian catchment appears worse than in the Welsh catchment, likely related to the fact that training samples for model development were taken from the western part of England. For application across the UK, it would be necessary to expand the number of training samples and include different regions.

The model also relies on the S2DR3 approach to downscale imagery from 10 m to 1 m resolution. Higher resolution aerial imagery would be effective in reducing false positives and misclassified pixels, although this would require additional training datasets and reconfiguration of model parameters, and would likely cost more to acquire.

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